



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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MORPHOLOGY AND WORD FORMATION IN NATURAL LANGUAGES

DSA0304 – Natural Language Processing

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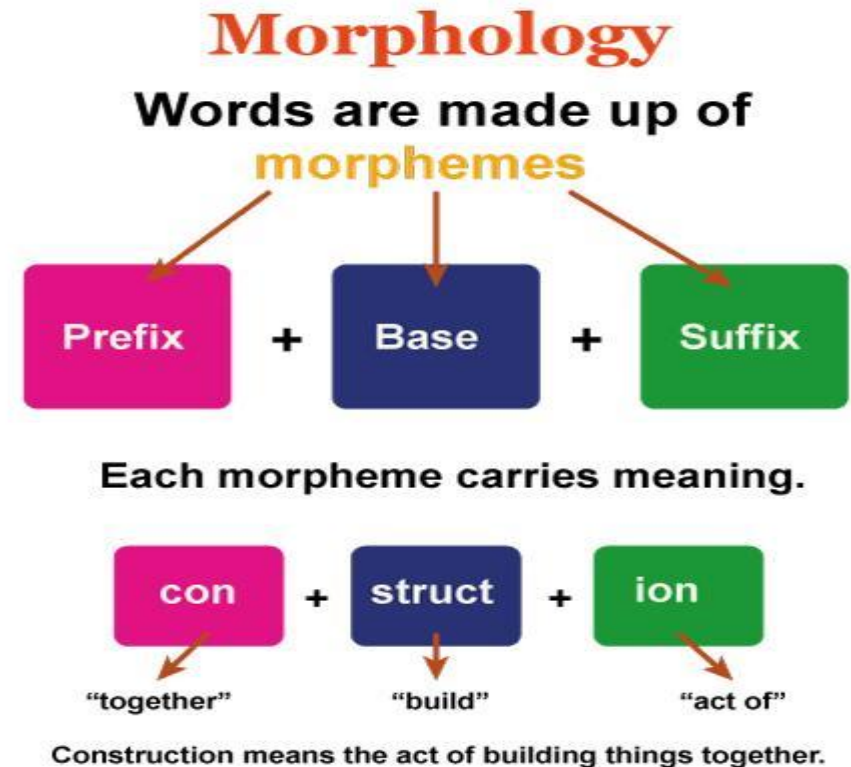
INTRODUCTION



- Morphology is the study of the internal structure of words and how new words are formed using the smallest meaningful units called **morphemes**.

➤ Key Objectives

- Understand word structure.
- Analyse prefixes and suffixes.
- Identify root words.
- Improve language understanding.
- Support Natural Language Processing applications.





MORPHEME



- A **morpheme** is the smallest meaningful unit of language.
- Morphemes play a vital role in word formation and vocabulary expansion.

Example

- Unhappiness

Part	Meaning
Un	Not
Happy	Root word
ness	State or Quality



TYPES OF MORPHEMES

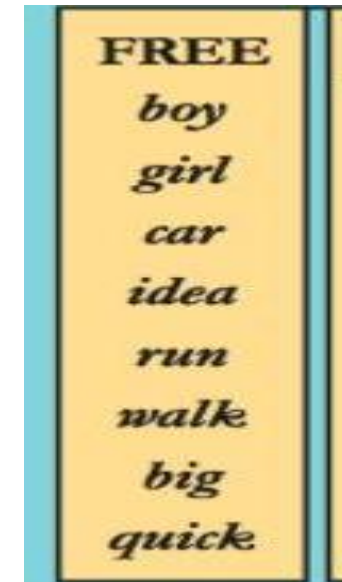


1.Free Morphemes

A Free Morpheme is a morpheme that can stand alone as an independent word and has a complete meaning without needing any additional morphemes.

- **Characteristics**
- Can function as a complete word.
- Carries lexical meaning.
- Can combine with bound morphemes to create new words.

Free Morpheme	Meaning
Book	A Collection Of pages
Run	To move Quickly
Happy	Feeling Joy



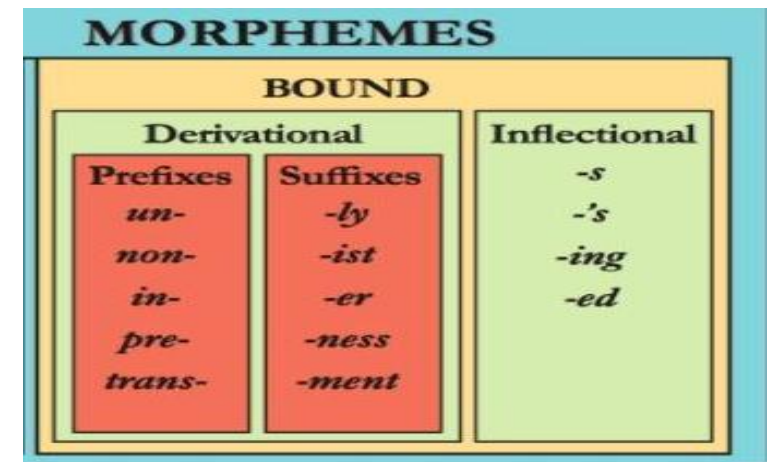


2. Bound Morphemes

A Bound Morpheme is a morpheme that cannot stand alone as a word. It must be attached to a free morpheme to express meaning or grammatical information.

- **Characteristics**
- Cannot exist independently.
- Usually appears as a **prefix** or **suffix**.
- Modifies the meaning or grammatical form of the root word.

Bound Morpheme	Function	Example
Re-	Again	rewrite
-ing	Continuous Action	Playing
-ed	Past Tense	Walked





TYPES OF WORD FORMATION



Process	Description	Example
Prefixation	Adding a prefix before the root word	happy → un happy
Suffixation	Adding a suffix after the root word	teach → teacher
Compounding	Combining two complete words	black + board = blackboard
Blending	Combining parts of two words	breakfast + lunch = brunch
Clipping	Shortening a longer word	examination → exam
Acronyms	Forming a word from initial letters	NASA, UNESCO



IMPLEMENTATION



```
technical_presentation.py > ...
1  # Morphology and Word Formation Implementation
2
3  word = input("Enter a word: ")
4
5  prefixes = ["un", "re", "dis", "pre"]
6  suffixes = ["ing", "ed", "ness", "ly", "er"]
7
8  prefix = ""
9  suffix = ""
10 root = word
11
12 # Check Prefix
13 for p in prefixes:
14     if word.startswith(p):
15         prefix = p
16         root = word[len(p):]
17         break
18
19 # Check Suffix
20 for s in suffixes:
21     if root.endswith(s):
22         suffix = s
23         root = root[:-len(s)]
24         break
25
26 print("\nMorphological Analysis")
27 if prefix:
28     print("Prefix :", prefix)
29 print("Root Word :", root)
30 if suffix:
31     print("Suffix :", suffix)
```



OUTPUT



```
● PS C:\NLP> python technical_presentation.py
Enter a word: unhappiness

Morphological Analysis
Prefix : un
Root Word : happi
Suffix : ness
❖❖ PS C:\NLP> █
```




CONCLUSION



- Morphology studies the structure and formation of words.
- Morphemes are the smallest meaningful units of language.
- Word formation expands the vocabulary of a language.
- Morphology helps NLP systems understand word meanings.
- It improves tasks like translation, spell checking, and text analysis.
- Morphology is a key foundation for modern Natural Language Processing.



Thank
you :-)